Day-39 23 May, 2025

DHCP

- DHCP allows DHCP clients, usually end devices like PCs, phones, etc. to automatically learn which IP address to use from a DHCP server, instead of us having to manually configure it.

Purpose of DHCP

- DHCP is used to allow hosts to automatically / dynamically learn various aspects of their network configuration, such as IP address, subnet mask, default gateway, DNS server, etc. without manual / static configuration

- It's an essential part of modern networks.
  ↳ When you connect a phone / laptop to WIFI, do you ask the network admin which IP address, subnet mask, default, etc. the phone / laptop should use?
  Probably not. Coz there's DHCP to automate it
- Typically used for 'client devices' such as workstations (PCs), phones etc.

- Devices such as routers, servers etc. are usually manually configured.

- In small networks, (such as home networks) the router typically acts as the DHCP server for hosts in LAN

- In larger networks, the DHCP server is usually a Windows / Linux Server.

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The basic functions of DHCP

- C:\Users\user >ipconfig /all.

DHCP enabled: Yes
IPv4 address: 192.168.0.167 (Preferred)
Lease Obtained: Lease Expired.

Preferred - It means if this PC was previously assigned this IP address by the DHCP server, so it asked to receive the same address again this time.

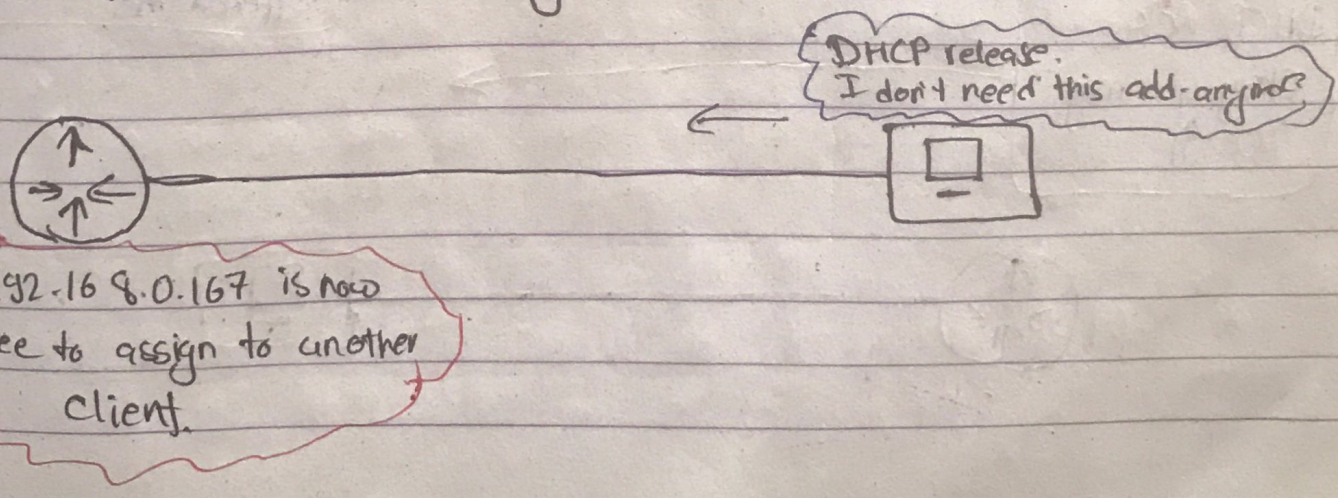
If the preferred address ain't available, device would be assigned different address.

Lease: DHCP server 'lease' IP address to clients. These clients are usually not permanent & client must give up address at end of lease. They can also release IP before lease, if they longer need it.

↳ because DHCP server can be configured to permanently assign IP addresses. But its best to provide lease.

To release IP address use the command.

C:\Users\user >ipconfig /release.



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NOB: DHCP Servers use UDP 67 (port)
DHCP Clients use UDP 68.

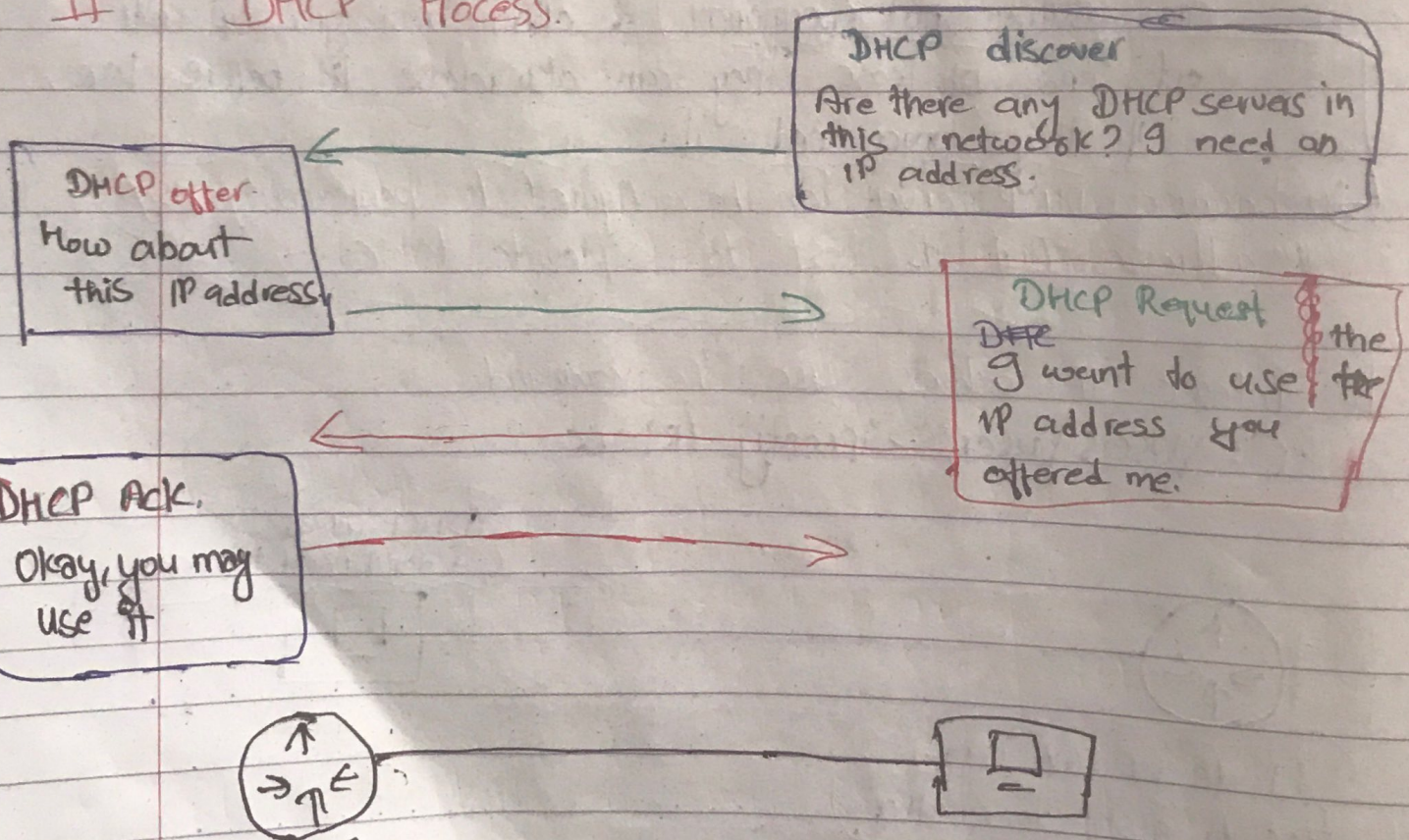
To get IP address from DHCP server.

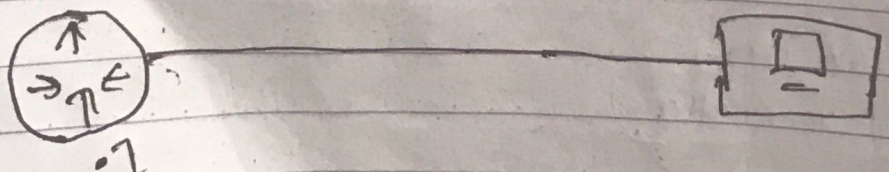
C:/Users/usa > ipconfig /renew

DHCP - Disover. From wireshark we can see:

- It sends broadcast addr as destination. & src is 0.0.0.0 indicating it doesn't have an IP add. yet.
- Bootp flags: 0x0000 (Unicast) - Bootp is predecessor of CCNA.

DHCP Process.



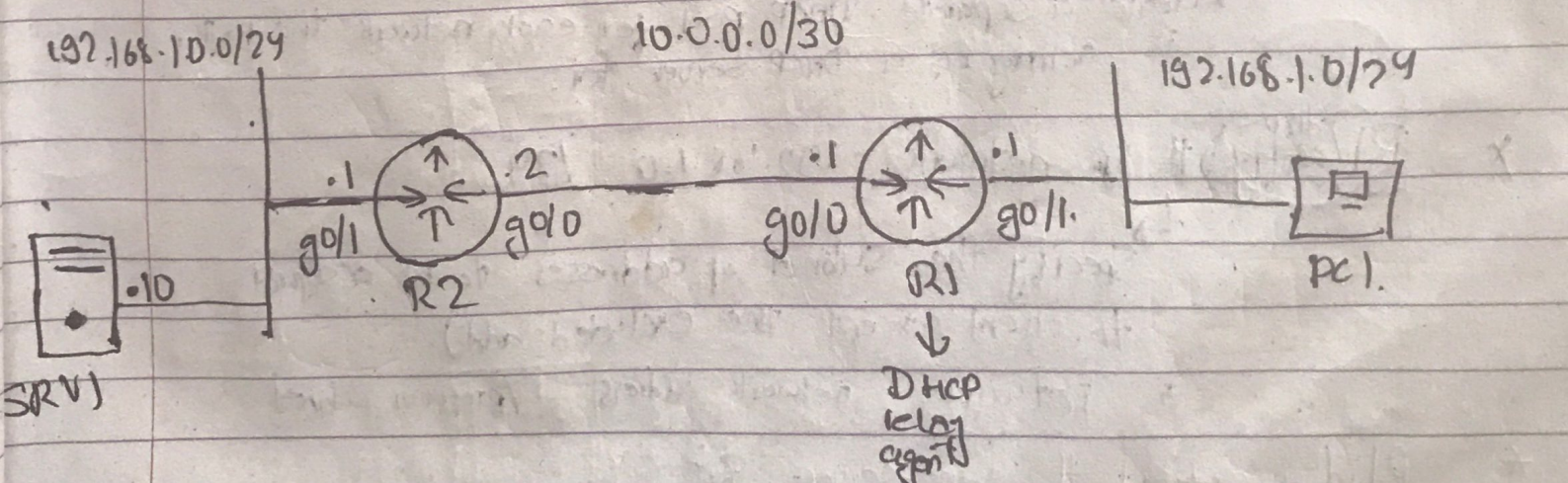


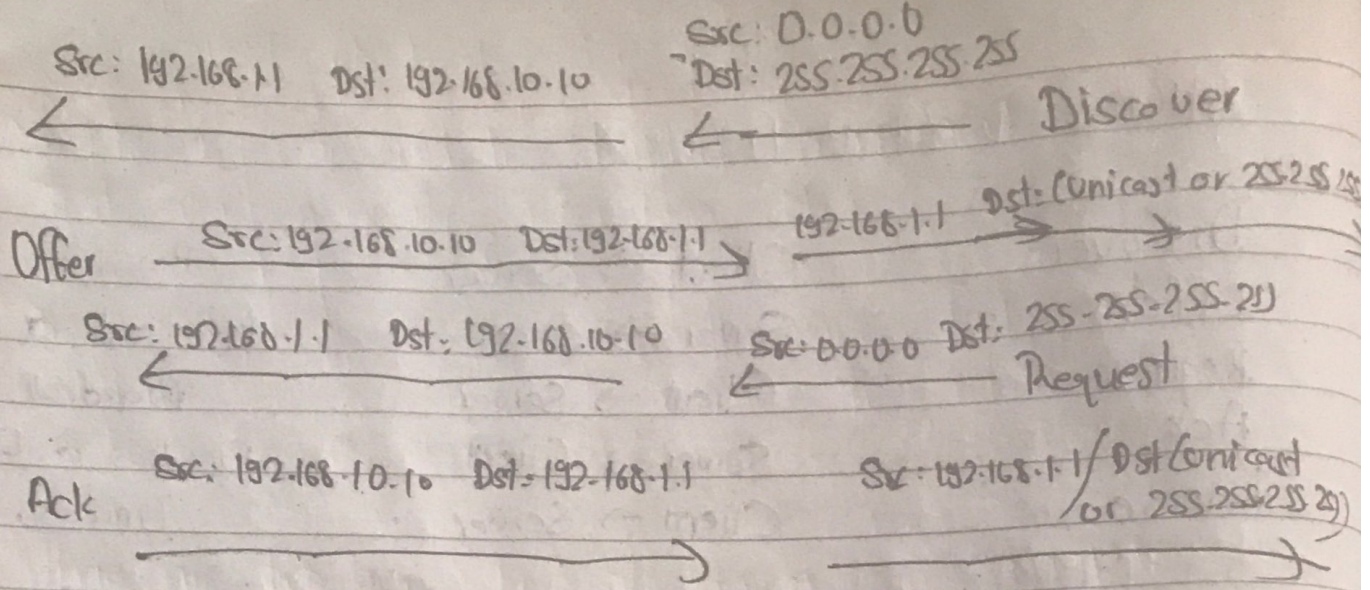
DHCP (D-O-R-A)

Discover	Client → Server	Broadcast
Offer	Server → Client	Broadcast or Unicast
Request	Client → Server	Broadcast
Ack	Server → Client	Broadcast or Unicast
Release	Client → Server	Unicast.

DHCP Relay

- Some network engineers might choose to configure each router to act as the DHCP server for its connected LANS
- However large enterprise often choose to use centralized DHCP server.
- If the server is centralized, it won't receive DHCP clients broadcast DHCP mssgs. (broadcast mssg, don't leave the local subnet)
- To fix this we can use configure, router to act as DHCP relay agent
- The router'll forward the clients broadcast DHCP mssg to remote DHCP server as unicast mssgs





DHCP Server Configuration in IOS

① CLI & PCI

* a) R1(config)#. ip dhcp excluded-address 192.168.1.1 192.168.1.10
   ↳ range of add. that won't be given to DHCP clients
   not necessary but better to reserve some for static configuration

* R1(config)# ip dhcp pool LAB_POOL (a name of our choice)
   ↳ this creates DHCP pool. gets subnet of addresses
   * that can be assigned to DHCP clients.
   * create a separate DHCP pool for each network the router is acting as a DHCP server for.

* R1(dhcp-config)# ip network 192.168.1.0 /24
   ↳ Specify the subnet of addresses to be assigned to client (except the excluded add).
   * network network-address /prefix or subnet

* R1(dhcp-config)# dns-server 8.8.8.8.
   specify the dns-server that clients should use.

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* R1(dhcp-config)# domain-name jeremysitlab.com
We can also specify domain-name of network.

* R1(dhcp-config)# default-router 192.168.1.1
* specifies default gateway

* R1(dhcp-config)# lease 0 5 30.
*specifies lease time.
lease days hours minutes OR
lease infinite (not recommended)

* R1# show ip dhcp binding.
It shows all of clients that are currently assigned IP addres

DHCP relay agent configuration

* R1 will be DHCP relay agent here. Agilai network hai

R1(config)# int g0/1 (* int connected to client-devices)
R1(config-if)# ip helper-address 192.168.10.10
ip heper-address ip-add-of-dhcp-server

DHCP client configuration in IOS

R2(config)# int g0/1
R2(config-if)# ip address dhcp
(* ip address dhcp dhcp - tells router to use DHCP to learn its IP address)